

Chapter 55: Socio-Economic Impacts of LPG Adoption Among Rural Women in Bihar: A Study Under Pradhan Mantri Ujjwala Yojana (PMUY)

Abhinaw kumar

Research scholar, Lovely Professional University

Email: pkumar94@hotmail.com

ORCID ID: 0009-0002-6410-5568

Abstract

Access to clean cooking energy remains one of the most pressing issues facing rural households in India. Bihar is among the most economically backward states in India, with a large share of its rural households still dependent on biomass fuels. This chapter examines the socio-economic impacts of LPG adoption on rural women in Bihar since the launch of PMUY, covering both the gains observed and the challenges that remain. It draws on primary data from rural households in six districts of Bihar, supplemented by secondary data from government reports and existing research.

Keywords: PMUY, LPG adoption, BPL, CWSEI

1. Introduction

Access to clean cooking energy remains one of the most pressing issues facing rural households in India. For decades, women in rural areas have cooked using traditional biomass fuels such as firewood, cow dung cakes, and agricultural waste, which cause serious health problems from indoor smoke and consume a large share of women's time each day.

Bihar is among the most economically backward states in India, with a large share of its rural households still dependent on biomass fuels. In 2016, the Government of India launched the Pradhan Mantri Ujjwala Yojana (PMUY) to provide free LPG connections to women in Below Poverty Line (BPL) households, aiming to transform cooking practices among the rural poor.

LPG produces far less smoke and harmful gases than firewood or dung cakes, benefiting health and freeing up time that women previously spent collecting fuel. However, an LPG connection alone does not guarantee regular use: many families revert to traditional fuels after the initial connection because of the cost of refills and limited awareness of the benefits of clean cooking fuel.

This chapter examines the socio-economic impacts of LPG adoption on rural women in Bihar since the launch of PMUY, covering both the gains observed and the challenges that remain. It draws on primary data from rural households in six districts of Bihar, supplemented by secondary data from government reports and existing research.

2. Background and Context

Energy poverty — the lack of access to modern, clean energy — remains widespread in rural Bihar. According to Census 2011, more than 88 percent of rural households in Bihar used firewood, chips, or dung cake as their primary cooking fuel, well above the national rural average of about 67 percent. Widespread unemployment, low literacy, weak infrastructure, and limited access to financial services all reinforce this dependence on biomass fuels. As the primary cooks in most households, rural women bear the heaviest burden, spending 3 to 5 hours a day collecting firewood and enduring long hours near a smoky traditional chulha.

The Government of India launched PMUY on 1 May 2016 in Ballia, Uttar Pradesh, under the Ministry of Petroleum and Natural Gas, to provide free LPG connections to BPL women. The scheme covers the security deposit, connection booklet, and cylinder (Rs. 1,600 per connection), with an EMI facility available for the first refill and stove. Bihar, with its large BPL population and very low pre-scheme LPG penetration, was a key target state, and connections rose steadily from 12.5 lakh in 2016–17 to a cumulative 85.2 lakh by 2021–22 (Table 1).

Year	Connections Released (Approx.)	Cumulative Total
2016–17	12.5 lakh	12.5 lakh
2017–18	18.3 lakh	30.8 lakh
2018–19	22.1 lakh	52.9 lakh
2019–20	14.7 lakh	67.6 lakh
2020–21	8.2 lakh	75.8 lakh
2021–22	9.4 lakh	85.2 lakh

Table 1: PMUY Connections Released in Bihar (Source: Ministry of Petroleum and Natural Gas, Annual Reports)

3. Review of Literature

International research links access to clean cooking fuel with women's well-being. WHO (2018) estimates that household air pollution from solid fuels causes about 3.8 million deaths a year globally. Duflo, Greenstone, and Hanna (2008) found that improved cookstoves in Orissa reduced indoor pollution, though the health gains depended heavily on actual usage. Bensch and Peters (2015) documented time savings for women in Senegal after switching to cleaner fuels, pointing to a pathway from energy access to economic empowerment.

In the Indian context, Gaikwad and Bhave (2019) found health improvements and more free time among PMUY beneficiaries in Maharashtra, while Kar and Sinha (2020) found low refill rates among BPL beneficiaries in eastern India, chiefly due to refill costs. Baidya et al. (2021) linked regular LPG use in Jharkhand and Bihar to greater self-confidence and household decision-making, while Mishra and Tripathi (2022) identified affordability, remote-area supply gaps, and low awareness as the principal barriers to sustained LPG use in Bihar, alongside caste-based and traditional social norms.

More recent work has deepened this picture. Vyas, Gupta, and Khalid (2021) found that patriarchal norms, more than infrastructure, sustain continued reliance on solid fuels. Chindarkar, Jain, and Mani (2021) identified affordability as the binding constraint on exclusive LPG use, and Su and Azam (2023) showed that LPG access reduces women's cooking time and improves employability without necessarily changing who does the cooking. Asharaf and Tol (2024) found PMUY's impact on LPG consumption modest and weaker in eastern states such as Bihar. Gaikwad, Pandey, and Patil (2024) reported continuing gaps in sustained fuel availability, and Singh, Pandey, and Kumar (2025) confirmed that patriarchal structures still constrain rural women's economic independence even where clean energy access has improved.

4. Theoretical Framework

The Energy Ladder Theory holds that as household income rises, families move from polluting fuels such as wood and dung towards cleaner ones such as LPG and electricity. In practice, however, many households do not progress linearly and instead "stack" fuels — for instance, using LPG for quick tasks while retaining firewood for rotis or festival cooking.

The Capability Approach, developed by Amartya Sen and elaborated by Martha Nussbaum, assesses development in terms of the real freedoms people can exercise rather than income alone. For rural women in Bihar, clean cooking fuel can expand the capability to live free of indoor air pollution, to direct time towards education and income-generating work, and to participate meaningfully in household and community decisions.

Drawing on these traditions, this study links LPG adoption to three categories of outcomes for rural women: economic (time use, income, expenditure), social (decision-making, status, children's education), and health/environmental (reduced indoor air pollution, respiratory illness, and injury). The framework also recognises that adoption is shaped by household income, awareness, refill infrastructure, government support, and prevailing social norms.

5. Research Methodology

The study was conducted across six districts of Bihar — Patna, Nalanda, Muzaffarpur, Sitamarhi, Gaya, and Nawada — purposively selected to represent Central, North, and South Bihar. A multi-stage stratified random sampling design was used: three blocks were randomly selected from each district by LPG penetration level (18 blocks total), four villages were drawn from each block using probability-proportional-to-size sampling (72 villages), and households were then selected using systematic random sampling from PMUY beneficiary lists. The final sample comprised 420 rural households, all PMUY connection holders, with the woman primarily responsible for cooking interviewed as the key respondent (Table 2).

District	Blocks	Villages	Households
Muzaffarpur	3	12	108
Sitamarhi	3	12	80
Patna	3	12	58
Nalanda	3	12	51
Gaya	3	12	83
Nawada	3	12	40
Total	18	72	420

Table 2: Sample Distribution by District

Primary data were collected using a pre-tested structured interview schedule covering household demographics, LPG use and refill patterns, time savings, fuel expenditure, self-reported health status, decision-making participation, livelihood and income changes, and barriers to LPG use. Focus Group Discussions were also conducted in 15 villages for qualitative insight. Data were analysed in SPSS using descriptive statistics, chi-square tests for associations between categorical variables, and regression analysis to identify determinants of adoption and refill frequency.

6. Socio-Economic Profile of the Study Area

The study area is socially diverse, predominantly Hindu with a significant Muslim minority, and includes Scheduled Caste, Other Backward Class, and General Category households. Average household size was 5.8 persons. Literacy stood at approximately 62 percent for men and 48 percent for women — both below the Bihar state average — and about 54 percent of women respondents had no formal education or only primary-level schooling.

Agriculture dominates local livelihoods: 58 percent of households depended on agricultural labour or cultivation, 22 percent on non-agricultural daily wage work, 12 percent on regular salaried employment, and the remainder on small trade, livestock rearing, and related activities (Table 3).

Occupation Category	Households	Percentage
Agricultural labour / cultivation	244	58.0%
Non-agricultural daily wage	92	22.0%
Regular salaried employment	50	12.0%
Small trade / business	21	5.0%
Others (livestock, etc.)	13	3.0%
Total	420	100.0%

Table 3: Occupational Distribution of Surveyed Households

7. LPG Adoption: Patterns and Trends in Bihar

All 420 respondents held PMUY connections, but actual usage varied considerably. Approximately 38.5 percent used LPG as their primary cooking fuel on a daily basis, 41.3 percent reported only occasional use — mainly for quick tasks or tea — and the remaining 20.2 percent had not refilled their cylinder in the preceding three months or more.

The average number of refills per year among active users was approximately 4.6, well below the national average of 8 to 10 refills for LPG-using households, confirming that PMUY beneficiaries in Bihar use LPG far less intensively than the scheme envisaged (Table 4).

Refills per Year	Respondents	Percentage
0 (no refill in last 3 months)	85	20.2%
1–3 refills	108	25.7%
4–6 refills	139	33.0%
7–9 refills	62	14.8%
10 or more refills	26	6.2%
Total	420	100.0%

Table 4: Distribution of Respondents by Annual LPG Refill Frequency

8. Economic Impacts of LPG Adoption

Time savings are among the most significant economic impacts of LPG adoption. Women reported spending an average of 3.2 hours a day collecting firewood and managing the chulha before receiving an LPG connection; this fell to about 0.4 hours a day for regular users afterward — an average daily saving of roughly 2.8 hours, or over 1,000 hours a year.

Women redirected this saved time in varied ways: about 20 percent used it for income-generating activities such as basket weaving, agricultural wage labour, tailoring, and dairy work; 40 percent spent more time helping children with studies; 24 percent used it for rest; 10 percent for community meetings and Self-Help Group activities; and 6 percent for other domestic activities (Figure 1).

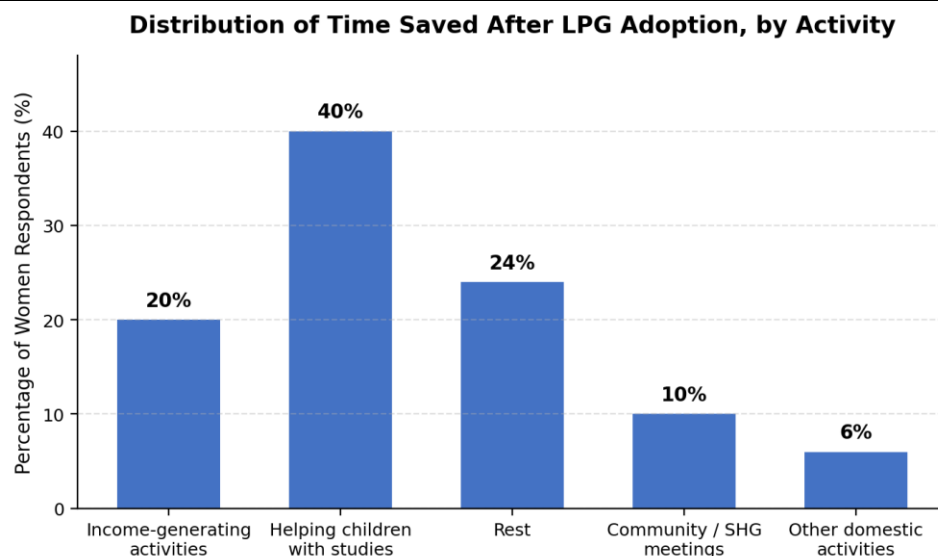


Figure 1: Distribution of Time Saved After LPG Adoption, by Activity (Field Survey Data)

Before PMUY, households collected firewood free of direct cost (at a high time cost) or purchased it for Rs. 450 to Rs. 600 a month. After LPG adoption, monthly refill costs run to approximately Rs. 700 to Rs. 850, a higher direct cash outlay that becomes more favourable once the value of time saved is factored in. Around 90 percent of respondents nonetheless identified refill cost as a significant household budget burden — the key driver of the low refill rates noted above. Among women who used saved time for income generation, average monthly income gains were Rs. 800 to Rs. 1,200, a meaningful addition for BPL families; women in agricultural labour in particular reported more working days once freed from early-morning firewood collection.

9. Social Impacts on Rural Women

LPG adoption improved the social standing of many women respondents. Regular users reported feeling more respected within their households, and focus group discussions revealed that women felt neighbours and relatives regarded them with greater respect for cooking on a "gas chulha," which many communities associate with modernity and economic progress.

About 44.3 percent of regular users reported improved participation in household decisions — on purchases, children's education, and agricultural matters — which women attributed both to more time for household discussion and to the practical confidence gained from managing the LPG connection. An important indirect effect was on children's education: girls previously expected to help collect firewood had more time for study, and smoke-free kitchens created a more comfortable space for children to sit and study. About 38 percent of respondents with school-going children reported improved school attendance, particularly among daughters (Table 5).

Social Indicator	Reporting Improvement
Improved social status / dignity	52.3%
Better household decision participation	44.3%
Improved children's school attendance	38.0%
More time for community / SHG meetings	29.7%
Reduced family conflict over fuel	33.5%

Table 5: Social Impacts of LPG Adoption Reported by Respondents

10. Health and Environmental Outcomes

Cooking with biomass fuels generates substantial indoor air pollution, with serious health consequences for women and young children who spend the most time near the cooking area. Regular LPG users reported significant reductions in self-reported health problems: 58.6 percent noted less cough and cold, 47.2 percent less eye irritation, and 42.5 percent improved breathing. About 34 percent had experienced a burn or fire-related injury while cooking with wood before switching to LPG, and reported such incidents falling markedly afterward.

While the study did not directly measure environmental indicators, key informant interviews with village leaders and forest department officials indicated a visible decline in firewood collection in the study areas since PMUY connections were provided, suggesting a positive environmental effect.

11. Women's Empowerment and Agency

Women's empowerment is understood here as a multidimensional concept spanning economic independence, freedom from drudgery, social recognition, decision-making participation, and the capacity for meaningful choice. The transition from biomass fuel to LPG is one dimension — though not the only one — of this broader process.

A Composite Women's Empowerment Index (CWSEI) was constructed from five indicators: economic participation, household decision-making, time autonomy, social mobility, and self-confidence, classifying women into Low, Moderate, and High empowerment categories (Table 6). Regular LPG users showed significantly higher empowerment levels than non-regular users, with the difference significant at the 1 percent level, confirming a meaningful positive association between sustained LPG use and women's agency.

Empowerment Level	Non-Regular Users (n=258)	Regular Users (n=162)
Low	52.8%	18.3%
Moderate	38.0%	51.5%
High	9.2%	30.2%

Table 6: Composite Women's Empowerment Index (CWSEI) by LPG Usage Pattern (Chi-square significant at $p < 0.01$)

12. Challenges and Barriers in LPG Adoption

Financial constraints are the most significant barrier: a 14.2 kg cylinder cost approximately Rs. 800 to Rs. 900 even after subsidy, a substantial expense for BPL households earning under Rs. 5,000 a month, with many able to afford a refill only once every two to three months. The PAHAL (DBTL) mechanism compounds this, since households must pay the full market price upfront and receive the subsidy only afterward, straining households with limited liquidity.

Supply and infrastructure gaps compound the problem: distributors in remote villages are often far from households, and transporting a cylinder by bicycle or on foot adds Rs. 30 to 70 to the effective cost, alongside irregular supply and long waits for home delivery. Social and cultural norms also play a role — some in-laws or husbands prefer the taste of chulha-cooked food, and in some households women are discouraged from handling the cylinder themselves, limiting their independent management of the connection. Awareness gaps persist too: about 44 percent of respondents could not explain how the subsidy is transferred, and 31 percent were unaware of current cylinder prices or how to check subsidy status.

13. Policy Implications and Recommendations

Reducing the effective cost of LPG for BPL households is the most urgent priority — through a higher per-cylinder subsidy for PMUY beneficiaries, the option of smaller 5 kg or 2 kg cylinders requiring lower upfront outlay, and interest-free loans or subsidised EMI facilities delivered through microfinance institutions and Self-Help Groups where formal banking access is limited.

Distribution infrastructure needs strengthening through new distributorships in underserved areas, mandatory home delivery at the distributor's cost, and regular monitoring to ensure PMUY beneficiaries receive cylinders promptly and without unofficial charges. Sustained awareness campaigns — delivered through Anganwadi and ASHA workers and SHG leaders who already hold community trust — should cover health benefits, the refill process, and subsidy entitlements. Community dialogues involving both men and women, alongside wider dissemination of success stories through radio and local media, can help shift attitudes around women's independent use of LPG. Finally, LPG access should be explicitly linked to broader livelihood programmes such as Jeevika, Mahila Samakhya, and Skill India, so that time saved is channelled into structured skill development and income generation.

14. Summary and Conclusions

This chapter has examined the socio-economic impacts of LPG adoption among rural women in Bihar, based on a sample of 420 PMUY-beneficiary households across six districts. LPG adoption under PMUY has had positive but incomplete impacts: while all sampled households held connections, only about 38.5 percent used LPG as their primary daily cooking fuel. Regular users gained an average of 2.8 hours a day, redirected substantially towards income generation, children's education, and community participation, alongside modest income gains of Rs. 800 to Rs. 1,200 a month for those taking up new livelihood activities.

Social impacts included improved dignity and household decision-making participation and better school attendance among children, particularly daughters, while health benefits included reductions in respiratory problems, eye irritation, and cooking-related injuries. The Composite Women's Empowerment Index confirmed a significant positive association between sustained LPG use and women's empowerment. The principal barriers to regular use were the cost of cylinder refills, weak supply infrastructure in remote areas, low awareness, and certain social-cultural norms.

LPG adoption under PMUY holds significant potential to improve rural women's lives in Bihar, but this potential remains far from fully realised because of affordability constraints and infrastructure gaps. Policy measures that make LPG more affordable, strengthen distribution, raise awareness, and integrate LPG access with broader livelihood programmes are essential to translate the scheme's intent into sustained, measurable improvements — a transition that represents far more than a change in cooking technology, but a step towards a more dignified, healthy, and empowered life for rural women.

References

- [1] Asharaf, N., & Tol, R. S. J. (2024). The impact of Pradhan Mantri Ujjwala Yojana on Indian households. *Economic Analysis and Policy*, 84, 878–897.

- [2] Baidya, S., Kumar, A., & Singh, R. (2021). LPG adoption and women's empowerment in rural Jharkhand and Bihar: A field study. *Journal of Rural Development*, 40(3), 215–234.
- [3] Bensch, G., & Peters, J. (2015). The intensive margin of technology adoption: Experimental evidence on improved cooking stoves in rural Senegal. *Journal of Health Economics*, 42, 44–63.
- [4] Chindarkar, N., Jain, A., & Mani, S. (2021). Examining the willingness-to-pay for exclusive use of LPG for cooking among rural households in India. *Energy Policy*, 150, 112127.
- [5] Duflo, E., Greenstone, M., & Hanna, R. (2008). Cooking stoves, indoor air pollution and respiratory health in rural Orissa. *Economic and Political Weekly*, 43(32), 71–76.
- [6] Gaikwad, H., Pandey, S., & Patil, K. (2024). Empowering rural Indian women by clean fuel initiatives: An impact assessment of Pradhan Mantri Ujjwala Yojana. *Cogent Social Sciences*, 10(1).
- [7] Gaikwad, P., & Bhawe, M. (2019). Ujjwala Yojana: Examining changes in health and time use among women beneficiaries in Maharashtra. *Indian Journal of Public Administration*, 65(2), 188–201.
- [8] Government of India. (2016). PMUY Scheme Guidelines. Ministry of Petroleum and Natural Gas. New Delhi: Government of India.
- [9] Government of India. (2022). Annual Report 2021–22. Ministry of Petroleum and Natural Gas. New Delhi: Government of India.
- [10] Kar, S., & Sinha, T. (2020). Determinants of LPG refill behaviour among PMUY beneficiaries in eastern India. *Energy for Sustainable Development*, 58, 87–98.
- [11] Mishra, V., & Tripathi, B. (2022). Barriers to sustained LPG adoption in rural Bihar: A multi-level analysis. *Economic and Political Weekly*, 57(14), 45–52.
- [12] Nussbaum, M. (2000). *Women and Human Development: The Capabilities Approach*. Cambridge: Cambridge University Press.
- [13] Sen, A. (1999). *Development as Freedom*. New York: Anchor Books.
- [14] Singh, J., Pandey, A. P., & Kumar, N. (2025). The impact of patriarchal norms on women's empowerment in rural Indian families. *Economic Sciences*, 21(1), 174–186.
- [15] Su, Q., & Azam, M. (2023). Does access to liquefied petroleum gas (LPG) reduce the household burden of women? Evidence from India. *Energy Economics*, 119, 106529.
- [16] Vyas, S., Gupta, A., & Khalid, N. (2021). Gender and LPG use after government intervention in rural north India. *World Development*, 148, 105682.
- [17] World Health Organization. (2018). Household air pollution and health. WHO Fact Sheet. Geneva: WHO.